

Last lecture we saw that if $v_1 \in \text{Span}\{v_2, v_3, \dots, v_n\}$ or equivalently

$$v_1 + \alpha_2 v_2 + \alpha_3 v_3 + \dots + \alpha_n v_n = 0 \quad (\text{linear dependence})$$

for some $\alpha_2, \dots, \alpha_n \in F$

then
$$\text{Span}\{v_2, \dots, v_n\} = \text{Span}\{v_1, v_2, \dots, v_n\}$$

In particular, we proved the following theorem

Theorem Let V be an F -vector space. If $v_1, v_2, \dots, v_n$ are linearly dependent vectors, where $v_i \neq 0$ then there is an integer $j \in \{1, 2, \dots, n\}$ such that

- (a) $v_j \in \text{Span}\{v_1, \dots, v_{j-1}\}$ and $v_1, \dots, v_{j-1}$ are F -linearly independent
- (b) $\text{Span}\{v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_n\} = \text{Span}\{v_1, v_2, \dots, v_n\}$

This observation tells us that Perhaps, one can find a linearly independent set $\{w_1, w_2, \dots, w_m\} \subseteq \{v_1, v_2, \dots, v_n\}$ such that

$$\text{Span}\{w_1, w_2, \dots, w_m\} = \text{Span}\{v_1, v_2, \dots, v_n\}.$$

We shall prove this assertion along with many other interesting facts in this lecture.

Theorem (A) Let V be a finite dimensional F -vector space and let $\{u_1, u_2, \dots, u_m\}$ be a linearly independent set of vectors. If

$$V = \text{Span}\{v_1, v_2, \dots, v_n\} \text{ then } \boxed{m \leq n}$$

Proof: Observe that each u_i is a nonzero vector (why?)

Since $u_1 \in V = \text{Span}\{v_1, \dots, v_n\}$, the set $\{u_1, v_1, v_2, \dots, v_n\}$ is linearly dependent (why?)

Now by (b), there is a $j \in \{1, 2, \dots, n\}$ such that

$$V = \text{Span}\{\underbrace{v_1, \dots, v_n}_{n\text{-vectors}}\} = \text{Span}\{\underbrace{u_1, v_1, \dots, v_n}_{n+1\text{ vectors}}\} = \text{Span}\{\underbrace{u_1, v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_n}_{n\text{-vectors}}\}$$

Next, we consider the linearly dependent set

$$\{u_1, u_2, v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_n\}$$

Again by (b), there is an $i \in \{1, 2, \dots, j-1, j+1, \dots, n\}$ such that

$$V = \text{Span}\{\underbrace{u_1, u_2, v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_n}_{n+1\text{ vectors}}\} = \text{Span}\{\underbrace{u_1, u_2, v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_{i-1}, v_{i+1}, \dots, v_n}_{n\text{-vectors}}\}$$

Now, we consider the linearly dependent set

$$\{u_1, u_2, u_3, v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_{i-1}, v_{i+1}, \dots, v_n\}$$

and repeat the above arguments to obtain that $\boxed{m \leq n}$

Note that if $m > n$ then at one point in the process, we will

have $V = \text{Span}\{u_1, u_2, \dots, u_n\}$ and that

$$u_{n+1} \in V = \text{Span}\{u_1, u_2, \dots, u_n\}$$

This Contradicts the assumption that $\{u_1, u_2, \dots, u_n\}$ is a linearly independent set.

Theorem Subspace of a finite dimensional vector space is finite dimensional.

Proof. Let V be a finite dimensional vector space and let $V = \text{Span}\{v_1, v_2, \dots, v_n\}$.

Let U be a Subspace of V .

- If $U = \{0\}$ then U is clearly finite dimensional.
- If $U \neq \{0\}$ then pick $0 \neq u_1 \in U$ and observe that $\{u_1\}$ is a linearly independent set
- If $U = \text{Span}\{u_1\}$ then U is finite dimensional. However, if $U \neq \text{Span}\{u_1\}$ then there is an element $u_2 \in U$ and $u_2 \notin \text{Span}\{u_1\}$. Now, $\{u_1, u_2\}$ is a linearly independent set.
- If $U = \text{Span}\{u_1, u_2\}$ then U is finite dimensional. However, if $U \neq \text{Span}\{u_1, u_2\}$ then there is an element $u_3 \in U$ and $u_3 \notin \text{Span}\{u_1, u_2\}$. Now $\{u_1, u_2, u_3\}$ is a linearly independent set

Proceeding like this, we observe that there must be an m such that $U = \text{Span}\{u_1, u_2, \dots, u_m\}$ (why?)

See the Previous theorem!

Thus U is finite dimensional.

Note:

If $\{u_1, u_2, \dots, u_m\}$ is a linearly independent set and $u_{m+1} \in V$ and $u_{m+1} \notin \text{Span}\{u_1, u_2, \dots, u_m\}$ then the set

$u_1, u_2, \dots, u_m, u_{m+1}$ is linearly independent.

Basis of a vector space.

A set $B = \{v_1, v_2, \dots\}$ of linearly independent vectors of V is called an F-Basis of V if $\text{Span}(B) = V$.

Examples

① The set $S = \left\{ \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} \right\}$ is a linearly independent set and that

$$\text{Span}(S) = \mathbb{F}^n$$

Therefore, S is a basis of $\mathbb{F}^n$.

② The set $\left\{\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\right\}$ is a basis for $\mathbb{F}^3$.

Proof: Since $\left\{\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right\}$ spans $\mathbb{F}^3$, we can have at most 3 linearly independent vectors in $\mathbb{F}^3$.

Now $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \notin \text{Span}\left\{\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}\right\}$ and therefore

$\left\{\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}\right\}$ is a linearly independent set.

Furthermore,

$\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \notin \text{Span}\left\{\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}\right\}$ and therefore

$\mathcal{B} := \left\{\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\right\}$ is a linearly independent set.

Now, if $v \in \mathbb{F}^3$ and $v \notin \text{Span}(\mathcal{B})$ then, as noted earlier,

$\left\{\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, v\right\}$ will become a linearly independent set, contradicting Theorem A.

(having 4 linearly independent vectors)

Thus, for any $v \in \mathbb{F}^3$, we must have $v \in \text{Span}(\mathcal{B})$.

Hence

$$V = \text{Span}(\mathcal{B})$$

and we obtain that $\mathcal{B}$ is a basis of V .

③ The vector space $\mathcal{P}$ of all polynomials over Real numbers has $\{1, x, x^2, \dots\}$ as a basis.

Proof Clearly,

$$\text{Span } \{1, x, x^2, \dots\} = \mathcal{P}.$$

Also, if $a_0 \cdot 1 + a_1 x + a_2 x^2 + \dots + a_n x^n = 0$
then, by definition, $a_0 = 0, a_1 = 0, \dots, a_n = 0$.

"A polynomial is zero if and only if all its coefficients are zero".

Thus $\{1, x, x^2, \dots\}$ is a linearly independent set.
Hence $\{1, x, x^2, \dots\}$ is a basis of $\mathcal{V}$.